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Five different perspectives on mathematical modeling in mathematics education

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ABSTRACT
This article presents a review of literature exploring five different perspectives on mathematical modeling in mathematics education. Because there is not a single agreed-on definition of what mathematical modeling is or how it should be done, a focused and extensive review of mathematical modeling is essential. The five broad classifications discussed in this article include realistic modeling, educational modeling, models and modeling perspective, socio-critical modeling, and epistemological modeling. For each perspective, we present (a) the goals of mathematical modeling, (b) the definition of a mathematical model, (c) the mathematical modeling cycle, (d) the design of the modeling task, and (e) key researchers and research foci. In this article, we aim to present the different perspectives of mathematical modeling in an organized way so as to (1) demonstrate the rich and diverse background of mathematical modeling and (2) clarify theoretical foundations of seminal works in mathematical modeling.

KEYWORDS
Mathematical modeling; mathematics education

Introduction

One of the standard explanations of why mathematics tends to get so much time and attention in formal educational settings involves the idea that mathematics is a useful tool to understand everyday life better, prepare for future employment, engage in intelligent citizenship, and make sense of other subjects that students are expected to learn (Pollak, 2007). In fact, researchers have explained that mathematics is necessary because it is an important part of culture and society, and it should be seen as a powerful tool for better understanding present and future real-world situations (Niss, 1996; Pollak, 2016). A natural question is "What might mathematics educators do to prepare all students to use mathematics for everyday life, future employment, and intelligent citizenship?" Since the 1960s, much work has been done in the field of mathematics education to understand how students engage in the application of mathematics to real-world contexts (Cirillo, Pelesko, Felton-Koestler, & Rubel, 2016). Generally, this process of using mathematics to solve real-world problems has come to be called mathematical modeling (Niss, Blum, & Galbraith, 2007; Pollak, 2016). Furthermore, Smith, Haarer, and Confrey (1997) conveyed how useful information about the real world can be found by modeling caricatures or thought experiments.

Beyond this general description, the process of teaching and learning mathematical modeling consists of a wide range of goals and backgrounds. For instance, Julie and Mudaly (2007) describe such differences by characterizing *modeling as a vehicle*—when modeling is used to develop

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mathematical understanding—or *modeling as content*—when modeling is used to develop the skills of modeling. The various perspectives on mathematical modeling can be attributed to the goals that are evident in what constitutes a mathematical modeling task, how the mathematical modeling process is described, and the purpose of the research or modeling itself. Although many researchers agree that mathematical modeling should be integrated into mathematics classrooms, a cohesive conceptualization of how that can and should be done has not been developed (Kaiser & Sriraman, 2006; Lesh & Fennewald, 2010). The different perspectives of mathematical modeling are influenced by the perceived goal(s) of the modeling process along with the theoretical beliefs of the researchers (Ferri, 2013). Such differences often lead to various definitions, frameworks, research foci, and implications. Some researchers (Blomhøj, 2009; Kaiser & Sriraman, 2006) have examined the various definitions and goals of mathematical modeling and developed broad classifications for the different perspectives of mathematical modeling.

Because there is not a single agreed-on definition of what mathematical modeling is or how it should be done, along with the rising awareness of the value of mathematical modeling as an important aspect of the mathematical experiences for each and every student, a focused and extensive review of mathematical modeling research is essential. In this article, we aim to present the different perspectives of mathematical modeling education in an organized way. The perspectives discussed in this article are limited to published research in the field of *mathematical modeling education*—research about the teaching and learning of mathematical modeling. We define each perspective, provide a review of the literature, and compare the five perspectives to determine similarities and differences to (1) demonstrate the rich and diverse background of mathematical modeling and (2) clarify theoretical foundations of seminal works in mathematical modeling.

Literature search procedures

In this review, the primary databases used were EBSCO Academic Search Premier, Education Resources Information Center (ERIC), ProQuest Research Library, and ProQuest Digital Dissertations. Additionally, sources were obtained through search engines (e.g., Google Scholar). Literature used in this review includes peer-reviewed journal articles, proceedings, and papers from national and international conferences, books, and dissertations.

Categorization of the different perspectives

The differences in the goals behind selecting specific mathematical modeling tasks serve as defining characteristics in the different perspectives. Niss et al. (2007), along with Julie and Mudaly (2007), categorize and examine these differences in goals as (1) facilitating the learning of mathematics and (2) exploring nonmathematical scenarios using mathematics as a tool. These classifications are not meant to represent a dichotomy; rather, they should be seen as a duality, because it is possible to work toward different levels of each goal at the same time (Niss et al., 2007).

The first category is described by Niss et al. (2007) as "modeling for the learning of mathematics" (p. 6), whereas Julie and Mudaly (2007) describe it as "modeling as a vehicle" (p. 504). The end goal of this category is for the learner to gain a deeper understanding of mathematics through the use of contextual situations. However, the end goals for the second category are not explicitly learning the mathematics itself, but rather the ability to go through the modeling process. Niss et al. (2007) describe this learning outcome as the application of mathematics to solve problems outside of mathematics. Conversely, Julie and Mudaly (2007) call this category "modeling as content" (p. 504) and state that modeling itself is the learning goal, without a specific focus on the mathematical content.

The following section is included to clarify the five different perspectives of mathematical modeling, because, as stated previously, there is not a single agreed-on definition of what mathematical modeling is, or how it is done. Rather, researchers have put forth different definitions, descriptions, and assumptions of what mathematical modeling entails (Cirillo et al., 2016). Accordingly, the subsequent section describes (a) the goals of mathematical modeling, (b) the definition of a mathematical model, (c) the mathematical modeling cycle, (d) the design of the modeling task, and (e) key researchers and research foci.

Five mathematical modeling perspectives

Along with the different purposes of mathematical modeling, the theoretical beliefs of the researcher play a key role in the varying perspectives on mathematical modeling (Ferri, 2013). Although many researchers agree that mathematical modeling has roots in the works of Piaget, Vygotsky, and pragmatists, other researchers have shown strong ethno-mathematics and socio-cultural influences in their works (Confrey & Maloney, 2006; Kaiser & Sriraman, 2006; Lesh & Doerr, 2003). Such influences create both subtle and foundational differences in the various frameworks and definitions of mathematical modeling.

Using the aforementioned categories of goals and varying theoretical backgrounds, Kaiser and Sriraman (2006) developed five broad classifications for the different perspectives in mathematical modeling education research, including realistic modeling, educational modeling, models and modeling perspective (or contextual modeling), socio-critical modeling, and epistemological modeling. These perspectives are not exclusive, in that many researchers have worked and published in multiple perspectives (Kaiser, Sriraman, Blomhøj, & Garcia, 2007). In the following subsections, we organized our synthesis of the literature using these five perspectives. The synthesis is summarized in Table 1.

Realistic modeling

When it comes to realistic modeling, the most defining aspect focuses on the goal of mathematical modeling, which is solving real-world problems. This perspective suggests that students must work with *realistic, authentic, and messy* tasks. Kaiser, Schwarz, and Buchholtz (2011) describe such tasks as "problems that are only a little simplified and… recognized by people working in this field as being a problem they might meet in their daily work" (p. 592). These authentic tasks should require students to go through the mathematical modeling cycle because one of the goals of this perspective is to support students' ability to apply the modeling process. The main goal of this perspective is that the learner will develop his or her ability to solve real-life problems through engaging in these authentic tasks (Kaiser & Sriraman, 2006).

The theoretical background of realistic modeling is based primarily on the perspective of Henry Pollak (1969, 1992, 2007, 2016) and emphasizes the value of solving real-life problems rather than focusing on the development of mathematical content (Blomhøj, 2009; Kaiser et al., 2011). Although this perspective emphasizes the application of the mathematical modeling process, there is not a single agreed-on cycle that researchers use. An example of a modeling cycle with roots in this perspective was developed by Blum (2011), which is a modified version of that of Blum and Leiss (2007). This mathematical modeling cycle has been adapted by various researchers within this perspective (Biccard & Wessels, 2011; Frejd & Ärlebäck, 2011; Ludwig & Reit, 2013). In this modeling cycle (shown in Figure 1), the modeler begins with the given *real situation and problem*. Subsequently, once the problem is understood, a *situation model* is constructed. Then, the modeler simplifies the problem by identifying key aspects or variables to structure a *real model*. This step is critical, because the original modeling task is presented in a *real, authentic, and messy* context. Next, *mathematization* occurs in which the real model is transformed into a *mathematical model* (such as an equation, a graph, etc.). This mathematical model allows the modeler to work with the mathematics and produce some *mathematical results*. Once the mathematical results are produced, they must be interpreted back into the real situation, which yields *real results*. Once the real results are

Table 1. Summary of the main features of the five mathematical modeling perspectives.

Perspective	Realistic	Educational	Models and modeling	Socio-critical	Epistemology
Goals	Develop skills to model and understand authentic, real-world scenarios.	Develop skills to model a real-world scenario and understand mathematics.	Develop a deep understanding of mathematics through a modeling context.	Develop mathematical modeling skills to make decisions in society.	Develop formal mathematical reasoning.
Definition of a mathematical model	Mathematical objects (graphs, equations, etc.) explain the given real-world scenario.	Mathematical objects that have a relationship to the given real-world scenario.	A conceptual system that maps the structural characteristics of a relevant system.	Mathematical representation of a relevant scenario.	The result of an activity based on situations and mathematical concepts.
Description of the mathematical modeling cycle	A cyclical, multistep process. Begins in the real world, is mathematized into the mathematical world, and ends in the real world.	A cyclical, multistep process that begins in the real world, is mathematized, then ends in the real world.	A cycle that begins in the real world, then, once a model is developed, it goes back to the real world. The cycle is repeated as many times as needed.	All aspects of the modeler's involvement in the exploration of a real-world problem using mathematics.	Four stages of activities. Models-of and models-for are created to develop formal mathematical reasoning.
Design of task	Authentic, messy, real-life tasks that require the use of the modeling cycle.	Authentic tasks that can be simplified to reveal specific mathematical goals.	MEAs designed to develop a specific mathematical concept. Must be in context of a real-world problem. Must meet 6 guiding principles.	Tasks are in societal context but should also focus on the development of specific mathematics concepts.	No set requirements.
Examples of key researchers	Pollak, Ferri	Niss, Blum, Zbiek, Conner	Lesh, Doerr	D'Ambrosio, Barbosa, Skovsmose	Freudenthal, Gravemeijer
Research focus	Mathematical modeling competencies	Mathematics in mathematical modeling and mathematical modeling in curricula.	The use of MEAs to teach mathematics.	Learners' use of mathematics to understand society critically.	Teaching and learning of specific mathematical concepts.

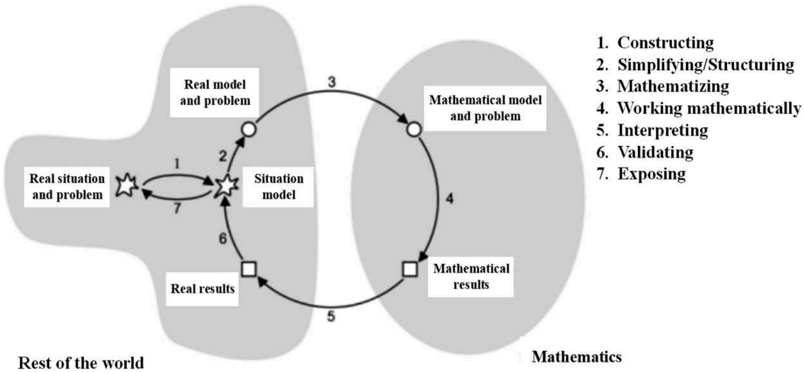

Figure 1. A modeling cycle from the realistic perspective.
(Adapted from Blum, 2011.)

produced, validation of these results must occur in order to report or expose the results of the given task, or the modeler goes back and creates another situation model (Blum, 2011).

Figure 1 shows mathematical modeling steps and subprocesses, where the steps result in products of the modeling cycle (such as a situation model, mathematical results, and real results) and the subprocesses explain the actions of the modeler during the modeling process (numbered 1–7 on the right side of the diagram). This allows the modeler's activities to be highlighted through each part of the modeling cycle (Blomhøj & Jensen, 2003).

Although researchers using this perspective may examine different educational topics, the main focus is on mathematical modeling competencies and tools that support mathematical modeling competencies (Blomhøj, 2009). Mathematical modeling competency is defined as being able to go through all the stages of a mathematical modeling process, independently and purposefully (Blomhøj & Jensen, 2003). Research studies that are framed within the realistic perspective tend to examine essential intricacies in mathematical modeling (Kaiser & Sriraman, 2006). In summary, realistic modeling emphasizes the development of modeling competencies used to understand authentic, real-world scenarios through mathematics.

Educational modeling

The goal of educational modeling is not only to develop mathematical modeling competencies (such as realistic modeling), but also to learn mathematics (Blomhøj, 2009). In the educational modeling perspective, both "modeling as content" and "modeling as a vehicle" (Julie & Mudaly, 2007) are emphasized simultaneously. Because this perspective highlights the relationship between the real world and mathematics as a vital element in the teaching and learning of mathematics, Kaiser and Sriraman (2006) described educational modeling as a continuation of Hans Freudenthal's late work (1973, 1981) in the field of mathematics education. The dual focus on mathematical modeling competencies and mathematics in educational modeling may result in the original tasks being more simplified than the messy authentic tasks described in the realistic modeling perspective. Furthermore, this view of modeling uses contexts to develop mathematical concepts, which leads to mathematical modeling. In other words, the goal of mathematical modeling tasks in this perspective is to motivate the need for learning mathematics (Kaiser et al., 2007).

Fundamental work within this framework includes that of Blum and Niss (1991). A mathematical model within this perspective is described as a *triple S.M.R.*, where the S stands for "some real problem situation," M stands for "a collection of mathematical entities," and R stands for "some relation by which objects and relations of S are related to objects and relations of M" (Blum & Niss, 1991, p. 39).

Additionally, mathematical models are created when the real problem is analyzed by the modelers relative to their interests, which leads to a real model that contains the necessary characteristics of the original real situation. Then, the real model is translated into mathematics, or mathematized. When the work is complete within the mathematics world and conclusions are drawn, they must be "re-translated" back to the real world to validate the model. This modeling process may require the modeler to go through the cycle several times (Blum & Niss, 1991). This process is very similar to the mathematical modeling process described in the realistic perspective, which ought to be expected because both share a common goal of advocating for the importance of modeling competencies. However, the two outlooks are separated by the notion that researchers within the educational perspective also examine mathematical learning, not only mathematical modeling.

Another seminal work in educational modeling includes research conducted by Zbiek and Conner (2006). In their study, the two researchers developed a "diagrammatic model of mathematical modeling as a process that allows for mathematical understandings to be identified as learners are engaged in modeling tasks" (Zbiek & Conner, 2006, p. 89). Their diagrammatic model (Figure 2) demonstrates how the modeler transitions between the two worlds, the real world and the mathematical one. The boxes in the diagram represent steps within the modeling cycle, and the two-direction arrows represent the subprocesses. The mathematical modeling process begins with the real world, where students explore the real-world situation they are given. In the real world, modelers *specify*—identify the conditions and assumptions (C&A) of the given scenario—and *observe mathematically*—examine what happens while exploring with mathematics. This includes identifying variables and constraints in the given situation and using mathematical concepts to describe the problem. Then, students identify mathematical ideas: properties and parameters (P&P) they can relate to the mathematical world or entity. This process is called *mathematizing* and results in moving from the real world to the mathematical entity. Once the modeler is in the mathematical world, *combining* takes place, in which properties and parameters are explained and justified to match the mathematical entity. At this point, modelers can also *analyze*—mathematically manipulate—and *associate*—connect to the real world. The last step is for the modeler to move back to the real world by going through three activities: (1) *highlighting*—describing clearly any unidentified properties and parameters on the mathematical entity; (2) *interpreting*—translating mathematical results into the real-world context; and (3) *examining*—validating conclusions with the goal of

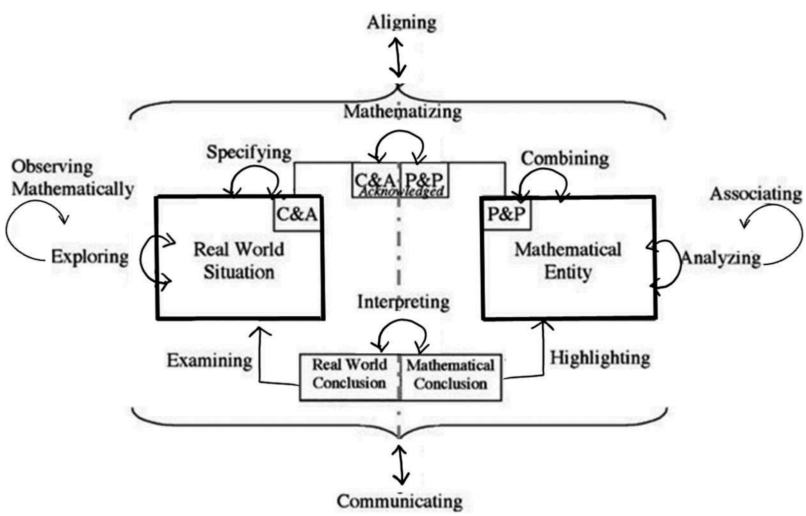

Figure 2. A modeling cycle from the educational perspective.
(Adapted from Zbiek & Conner, 2006)

the task. Throughout the entire process, the modelers are expected to apply the two subprocesses of *aligning*—constantly comparing the current state to the former ones—and *communicating*—putting forth ideas about the task between modelers (Zbiek & Conner, 2006).

Comparing the two mathematical cycles described above (Figures 1 and 2), one can see that both begin in the real world, mathematizing takes place to create a mathematical model, a mathematical solution is found and translated back into the real world and then validated. This perspective is also similar to the realistic perspective discussed above in that the mathematical modeling task should be authentic and realistic.

The research agenda in this perspective varies in that some researchers examine various approaches to mathematical modeling (Blomhøj, 2009), while others examine modelers' mathematical understanding while engaging in mathematical modeling tasks (Zbiek & Conner, 2006). In the educational modeling perspective, researchers focus on both modeling competencies and mathematical learning (Kaiser & Sriraman, 2006).

Models and modeling perspective (MMP)/contextual modeling

The following mathematical modeling perspective is called *contextual perspective* by Kaiser and Sriraman (2006) and *models and modeling perspective* (MMP) by Lesh and Doerr (2003). Additionally, it is also called model-eliciting perspective by Kaiser and Sriraman (2006). Nonetheless, this perspective describes a mathematical modeling framework that was developed by Lesh and colleagues; accordingly, we refer to this perspective as the MMP from this point on.

The MMP adapts the goal of problem solving to encompass the use of mathematical concepts and developing a deeper meaning for those concepts (Lesh & Doerr, 2003). This reveals an overlap with the goal of modeling in the educational perspectives, because modeling in this perspective is also viewed as content as well as a vehicle. However, there is a stronger emphasis on the learning of mathematics (Kaiser & Sriraman, 2006). The fundamental difference between the MMP and the educational perspective is the emphasis on model-eliciting activities (MEAs) and what constitutes a mathematical model. The MMP links mathematical modeling, and the learning that occurs through it, strongly to learning theories. Lesh and his colleagues' research (Lesh & Doerr, 2003; Lesh, Hoover, Hole, Kelly, & Post, 2000) have been influenced by the work of Piaget, and this influence is represented in their description of the mathematical modeling process and definition of models as

> … conceptual systems (consisting of elements, relations, operations, and rules governing interactions) that are expressed using external notation systems, and that are used to construct, describe, or explain the behaviors of other system(s)—perhaps so that the other system can be manipulated or predicted intelligently… [and a mathematical model] focuses on structural characteristics of the relevant systems (Lesh & Doerr, 2003, p. 10).

This definition differs from the previously discussed perspectives in that both the realistic and the educational perspectives view mathematical models as mathematizations of the real-world problem, while in this perspective, a mathematical model is described as a conceptual tool of a mathematical system that develops from a specific real-world situation (Lesh, Doerr, Carmona, & Hjalmarson, 2003). Rather than viewing mathematical models as mathematical objects relating to a given real-world scenario as educational modeling does, the MMP views them as systems, composed of different concepts, that map different characteristics of related systems.

Model-eliciting activities

As stated previously, in the MMP there is a strong emphasis on MEAs. To construct mathematical models, as they are defined above, the real-world tasks in this perspective are designed specifically to promote mathematical development. These real-world tasks are called model-eliciting activities (MEAs). MEAs are purposefully tailored around incorporating the mathematics process in the produced model. This serves as a major difference from mathematical modeling tasks within the realistic and education perspectives, because the goal of the design of an MEA is that, through engaging in the task, the constructed model

Table 2. Six guiding principles of MEAs (adapted from Lesh et al., 2000).

Principle	Description of the principle's requirements
(1) Model construction	Explicit construction, description, explanation, and justified predictions are required. The modeler should be able to quantify, coordinate, make predictions, and identify patterns or trends.
(2) Reality	Meaningful and relevant problems. Tasks must be based on real or slightly modified real data. Modelers must function as scientists or engineers who are working to solve a specific problem.
(3) Self-assessment	The task should promote selection, refinement, and elaboration of a model through a clear purpose and clear criteria of when a solution is achieved. Students should be able to detect deficiencies, compare alternatives, integrate strengths, and minimize weaknesses through assessing adaptions of models.
(4) Model documentation	Students must be able to reveal their thinking processes within their solution through descriptions of their assumptions, goals, and solution paths. Metacognition is supported in this principle.
(5) Generalizability	The model should represent a general way of thinking instead of specific solutions for a specific context. Models should be communicated in a clear and understandable manner so that they can be used by others.
(6) Effective prototype	The model must be simple but powerful. The model should avoid the need for numerous procedures, especially those that can circumvent conceptual understanding of mathematical topics.

represents the mathematical process (Kaiser & Sriraman, 2006). As a result of the important implications of MEAs, Lesh and colleagues developed six guiding principles for designing them. These principles (which are summarized in Table 2) include (1) the model construction principle, (2) the reality principle, (3) the self-assessment principle, (4) the model documentation principle, (5) the generalizability principle, and (6) the effective prototype principle (Lesh et al., 2000).

According to the *model construction principle*, the task should require modelers to describe how they develop their mathematical model. The *reality principle* requires the task to be authentic and take place in the real world. The authenticity of MEAs differs from the way authentic tasks are described in the realistic and educational perspectives, in that the problem does not have to be "real in an absolute sense" (Lesh et al., 2000, p. 616). Rather, tasks need to be set in the real world but not necessarily be a problem or situation that the modelers encounter in their daily life. The *self-assessment principle* requires the modelers to validate and be critics of their own work. Similar to the model construction principle, the *model documentation principle* requires the modelers to produce somewhat of an audit trail of their work. The *generalizability principle* asks the modelers to form a model that is open enough that it can be applied, or generalized, to similar situations. Lastly, the *effective prototype principle* requires the constructed model to be as simple as possible yet be a powerful mathematical tool to understand a complex situation.

The modeling cycle in this perspective is iterative, similar to those discussed previously in the realistic and educational perspectives. Lesh and Doerr (2003) described this iterative process as beginning in the real (or imagined) world, and through description, a model is constructed. Then, through manipulation of the model, a possible solution to the original task is achieved. Through prediction, the results are translated back into the real world, where verification takes place to confirm the solution. This process is depicted in Figure 3.

Research within the MMP requires the use of model-eliciting activities with a focus on the use of MEAs to develop and advance mathematical reasoning and understanding (Kaiser & Sriraman, 2006). Doerr and Lesh (2011) explained that the use of MEAs is critical to investigate learners' mathematical thinking. In fact, Lesh et al. (2000) referred to MEAs as thought-revealing activities and described them as being useful—to "learn more about the nature of students' (or teachers') developing knowledge, it is productive to focus on tasks in which the products that are generated reveal significant information about the ways of thinking that produced them" (p. 592). The research focus in this perspective differs from the educational perspective because modeling competencies are not of major concern, they are only an inherent aspect of the perspective (Blomhøj, 2009).

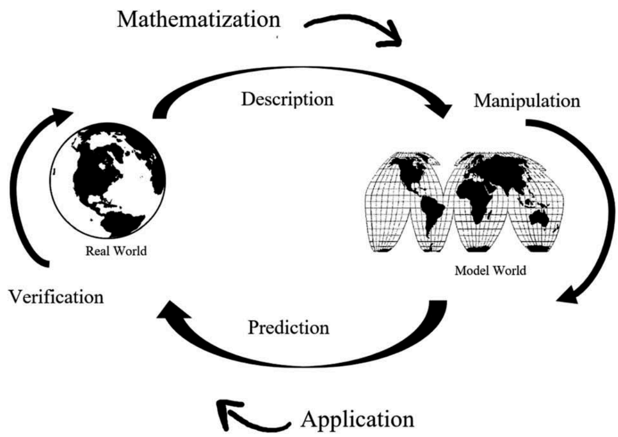

Figure 3. A modeling cycle from the MMP.
(Adapted from Lesh & Doerr, 2003)

Socio-critical modeling

The socio-critical modeling perspective focuses on the societal role that mathematics can play. This can be described as *modeling as critic* (Barbosa, 2006), with the goal of using mathematical concepts to make decisions in society. The goals of this perspective lie inside and outside of mathematics. The main goal within mathematics is to demonstrate the power of mathematics in making decisions and how it can be used as a tool to make decisions. However, the end goal of this perspective is to make decisions about society and to understand the mathematical modeling process critically (Kaiser et al., 2007).

The foundation of this perspective is built on the work of Skovsmose (1994; Skovsmose & Borba, 2004) in the field of critical mathematics pedagogy, in which two main objectives arise. First, critiquing what is being learned is an important part of attaining knowledge, and second, mathematics can actively influence decision making in society (Barbosa, 2006). Accordingly, Barbosa advocates for the learning of mathematics and mathematical modeling in a manner that satisfies both objectives. He explains that mathematical models are not neutral in nature, because they depend on their development process, and the development process depends on how the modelers see and understand the problem-situation, as well as how they adapt mathematical concepts into the situation. Therefore, this biased nature of mathematical modeling creates an authentic relationship between the mathematical model and the conditions applied to develop it (Barbosa, 2009). Additionally, this perspective brings up the idea of power behind mathematics: Because mathematical modeling is used to make decisions in society, modelers who are able to both create and critique mathematical models hold the power. Therefore, this perspective promotes mathematical modeling for students to develop the skills to become independent citizens (Blomhøj, 2009).

Furthermore, mathematical modeling in this perspective is defined as "a learning milieu" in which students investigate a real-world problem via mathematics (Barbosa, 2006, p. 294). The modeling process is defined as a school activity that is taken from everyday contexts (Barbosa, 2006), which is a much broader definition than how the other perspectives describe the modeling process. This perspective does not describe how the modeling cycle takes place; it simply describes the mathematical model as a mathematical representation of a presented real-world situation (Barbosa, 2007).

Research studies in this perspective tend to focus on how students can apply mathematics critically to make sense of the world (Ferri, 2013). This differs from the realistic and educational modeling perspectives because mathematical competencies are not emphasized. It differs from the MMP because the focus is not on mathematical understanding. Rather, research within the socio-critical modeling perspective centers around students' ability to be critical modelers and recognize their power.

Epistemological modeling

The epistemological perspective of mathematical modeling perceives mathematical modeling as a lens to establish general theories for the teaching and learning of mathematics (Blomhøj, 2009). The only goal for this perspective is the development of mathematical understanding. In fact, unlike any of the other perspectives discussed above, this perspective does not require a real-world problem or situation to be modeled. Kaiser and Sriraman (2006) explain that in the epistemological perspective, every mathematical task can be described as a modeling task, and modeling is not limited to the mathematizing of nonmathematics problems. However, modeling activities in this perspective often begin in the real world but end in mathematics (Kaiser & Sriraman, 2006).

The learning theories that underlie the research within this perspective are another distinctive feature of epistemological modeling. The *realistic mathematics education* (RME) theory is often used to conduct research within this perspective (Blomhøj, 2009; Kaiser et al., 2007). Accordingly, the rest of this section will describe mathematical modeling in the context of RME.

Emergent modeling

Emergent modeling was developed by Koeno Gravemeijer and his colleagues through the RME learning theory (Kaiser et al., 2007). The central goal for this type of modeling is focused on the learning process. The initial model that is developed by the learner is founded through an experience, and over time, the learner is able to recognize the model as it transitions from an informal experience to formal mathematical understanding (Gravemeijer, 2007). This outlook is rooted in the idea that mathematics should be connected to the learners' experiences and embedded in human activity (Doorman & Gravemeijer, 2009). Moreover, Gravemeijer (2002) explained that the emergent modeling perspective was established as a result of the application of external representations in mathematics education that later develop intrinsic meaning.

Gravemeijer (2002, 2007) described the modeling process in this perspective as containing four stages: (1) activity in the task setting, (2) referential activity, (3) general activity, and (4) formal mathematical reasoning. In the first stage, the learner is experiencing the task. Then, in the second stage, the learner develops "models-of" the specific context of the task. Next, in the general activity stage, the learner creates "models-for," which are mathematical entities, and these models are not connected to the context like the "models-of" in the referential activity stage. In the formal mathematical reasoning stage, the "models-for" are no longer needed by the learner, and formal mathematical understanding emerges.

Generally, researchers that apply the emergent modeling perspectives tend to investigate the teaching and learning of mathematical topics, such as calculus (Gravemeijer & Doorman, 1999; Doorman & Gravemeijer, 2009), data analysis (Gravemeijer, 2002), and functions (Doorman, Drijvers, Gravemeijer, Boon, & Reed, 2012), through the design of teaching experiments.

Synthesis of the five perspectives of mathematical modeling

The mathematical modeling perspectives discussed above reveal that the mathematical modeling education field has a diverse and extensive background. Even though the perspectives vary in many key aspects they also share some common ground. Table 1 displays a synthesis of the main features of each perspective in terms of its goals, its definition of a mathematical model, its description of the mathematical modeling cycle, the design of the modeling task, key researchers, and its research focus. The development of the table was guided by the works of Kaiser and Sriraman (2006) and Blomhøj (2009). These perspectives are foundational to many research studies in the field of mathematical modeling education and mathematics education in general.

Concluding remarks

Throughout the years, many researchers have described how mathematical modeling can be used as a tool to develop mathematical competencies that help students solve problems arising in everyday life, society, and the workplace (Lesh et al., 2000; Blum & Ferri 2009; Niss, 1996; Zbiek, 1998), and how mathematical modeling provides a venue through which learners can engage in mathematics in robust ways (Zbiek & Conner, 2006). However, the Program for International Student Assessment (PISA), which assesses education systems by testing the skills and knowledge of students worldwide, has indicated that students struggle in solving mathematical modeling tasks (OECD, 2012). Thus, along with the implementation of the Common Core State Standards, mathematical modeling has been brought to the forefront of learning and teaching mathematics. Because there is not a single agreed-on definition of what mathematical modeling is or how it should be done, along with the rising awareness of the value of mathematical modeling as an important aspect of the mathematical experiences for each and every student, a focused and extensive review of mathematical modeling is essential. As stated previously, this article focused on the different perspectives of mathematical modeling education, and thus it presents an incomplete and still developing picture of the nature of mathematical modeling. Accordingly, future work should include the perspectives of applied mathematicians, mathematical scientists, professional engineers, or other professionals who often practice mathematical modeling.

The goal of this article was to present a synthesis of the different perspectives of mathematical modeling in mathematics education in an organized way. We defined each perspective, provided a review of the literature, and compared the five perspectives to determine similarities and differences to (1) demonstrate the rich and diverse background of mathematical modeling and (2) clarify theoretical foundations of seminal works in mathematical modeling. It is our hope that our work will help the mathematics education community better understand mathematical modeling as well as mathematics learning through mathematical modeling.

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